

Bring Your Own Data

Block 4 — Modern Workflow in Geology

Ondrej Lexa

Institute of Petrology and Structural Geology (IPSG)
Charles University, Prague, Czechia

8 October 2026

Objective

Load and analyse data from your own thesis — with time for consultations and debugging.

`04a_byod_structural.ipynb`

- Field measurements (CSV/Excel, long or wide table)
- Stereonet per structure + summary with fold axis
- Statistics table; optional fault stress inversion

`04b_byod_epma.ipynb`

- Microprobe export (Excel/CSV)
- APFU, end-members, QC per mineral
- Ternary/scatter/profile charts, one results workbook

Using the Templates

Three Steps

1. **Copy** the template (keep the original as reference) and your data file into notebooks/data/
2. Edit **only the CONFIG cell**: file name, sheet, column names, which structures/minerals to process
3. *Run* → *Run All Cells* and look in notebooks/output/

Golden rule

Never edit the raw data file to “make it work”. Fix things in code (RENAME, SKIPROWS, filters) so the path from raw data to figure stays reproducible.

Structural Template — CONFIG

```
PROJECT = "field" # used in output file names
FILE = Path("data/field_measurements.csv") # .csv, .xlsx or .xls
LAYOUT = "long" # "long": structure column + azi, inc | "wide": S1_azi, S1_inc, ...
STRUCTURE_COL = "structure"
AZI_COL, INC_COL = "azi", "inc"
PLANE_NOTATION = "dd" # "dd" dip direction/dip | "rhr" right-hand-rule strike/dip
PLANES = ["S0", "S1"]
LINES = ["L1"]
FOLDED = "S0" # surface whose pi-girdle gives the fold axis (or None)
FAULTS = Path("data/mele.csv") # fazi, finc, lazi, linc, sense (or None)
FAULT_SENSE_LETTERS = False # True if sense is coded n/r/s/d instead of 1/-1
```

Output: <PROJECT>_<structure>.pdf, _summary.pdf, _faults_*.pdf, _apsg.xlsx
(statistics, data, stress inversion)

EPMA Template — CONFIG

```
SAMPLE = "S12"
FILE = Path("data/epma_export.xlsx")
SHEET = "Spots"
SKIPROWS = 0 # header lines above the column names
RENAME = {} # e.g. {"SiO2(Mass%)": "SiO2"}
MINERAL_COLUMN = None # a column that already holds the mineral ...
MINERAL_FROM = ("Comment", r"_([a-z]+)_") # ... or a regex applied to a column
MINERALS = {"grt": (Grt, (98.5, 101.5)), # label -> (mineral, total window)
            "pl": (Fsp, (98.5, 101.5)),
            "bt": (Bt, (94.0, 97.5))}
MIN_QUALITY = 0.95 # stoichiometry_quality threshold
PROFILE = ("Grt-profile", "Distance_um") # optional traverse sheet, or None
```

Output: QC summary, <SAMPLE>_*.pdf charts, <SAMPLE>_petropandas.xlsx

Common Pitfalls

Symptom	Likely cause / fix
Planes plot 90° off	strike/dip given as dip direction/dip — set <code>PLANE_NOTATION = "rhr"</code> (or quadrant strings: <code>fol("N30E,40NW")</code>)
Poles where planes should be	plotted with <code>s.line</code> instead of <code>s.pole</code> / <code>s.great_circle</code>
Fold axis nonsense	mixed limbs of several fold generations, or a cluster instead of a girdle — check P/G in the statistics table
Stress inversion unstable	sense coding: 1 normal, -1 reverse (letters n/r/s/d: set <code>FAULT_SENSE_LETTERS = True</code>); mixed fault populations — filter by misfit
Rows silently missing	empty cells — accessors skip rows with NaN

Symptom	Likely cause / fix
Oxide missing from <code>df.oxides()</code>	odd header (<code>SiO2 wt%</code> , <code>SiO2(Mass%)</code>) — add to <code>RENAME</code> ; <code>FeO*</code> , <code>FeOT</code> , <code>H2O+</code> work as-is
Header is garbage	instrument writes lines above the table — set <code>SKIPROWS</code>
Numbers read as text	decimal comma or “<LOD” strings — <code>pd.read_csv(..., decimal=",")</code> , <code>pd.to_numeric(errors="coerce")</code>
Fe^{3+} looks wrong	both <code>FeO</code> and <code>Fe2O3</code> present, or wrong mineral object (<code>Grt</code> vs. <code>GrtFe3</code>)
Everything rejected	total window too tight for hydrous minerals, or wrong mineral label mapping in <code>MINERALS</code>

Wrap-up

- **Documentation & tutorial notebooks**
 - github.com/ondrolexa/apsg
 - github.com/ondrolexa/petropandas
- **Keep your project reproducible:** one uv project per thesis, raw data read-only, notebooks under version control (git)
- **Grow the templates:** they are ordinary notebooks — add cells, cite the code in your methods section
- **Bugs and wishes:** GitHub issues on either repository

Thank you — now, your data!